Dublin Housing Affordability Analysis (2015–2025)

Chandini Konduru (25143913), Singireddy Sneha (24285854), Kapa Adheeshwar Reddy (25122932)
National College of Ireland
MSc in Data Analytics

***Abstract*—Housing affordability is one of the most pressing urban policy concerns in Dublin because property prices have risen rapidly while household purchasing power has not always kept pace. This project investigates whether housing in Dublin became less affordable relative to income over the period 2015–2025, and whether affordability pressure is distributed evenly across local areas. To answer this question, we designed and implemented an end-to-end analytics system using multiple public datasets and two database technologies.**

The project integrates three non-Kaggle data sources: (i) the Irish Property Price Register (transaction-level CSV), (ii) the Central Statistics Office (CSO) income dataset via JSON-stat API, and (iii) the CSO population dataset via JSON-stat API as contextual enrichment. The workflow was implemented in Python and organized as a reproducible ETL architecture. Raw records and payloads were persisted in MongoDB to preserve data lineage. Curated outputs were loaded into PostgreSQL for structured querying and dashboard-ready serving. Data processing included schema normalization, temporal filtering, quality checks, aggregation, and the derivation of affordability metrics.

Our principal indicator is an affordability index defined as average annual Dublin property transaction price divided by annual income. Results indicate a clear long-run affordability deterioration over available overlap years, with property prices increasing materially faster than income. Spatial analysis also reveals sustained area-level inequality: several high-cost areas remain persistently expensive across multiple years rather than exhibiting isolated volatility. Outputs were communicated through static analysis plots and a Streamlit dashboard combining KPI cards, trend charts, ranked area bars, heatmaps, and textual insights.

I. Introduction

Housing affordability has become a structural challenge in many cities where demand growth, constrained supply, and macroeconomic uncertainty interact over long periods. Dublin is a high-relevance case because the market combines strong economic activity with significant pressure on available housing stock. In such environments, a simple upward trend in prices is informative but incomplete. Affordability is a relational concept: it depends on the ratio between housing costs and household income, and is further shaped by local variation in neighborhood-level price dynamics.

This project addresses the following research question:

> Is housing in Dublin becoming unaffordable relative to income over time (2015–2025), and how does this affordability pressure vary across areas?

The project objectives were:

1) Build a reproducible analytics pipeline integrating structured and semi-structured datasets.
2) Programmatically store raw and processed data in two different databases (MongoDB and PostgreSQL).
3) Engineer annual affordability metrics and area-level comparisons for Dublin (2015–2025).
4) Produce both static and interactive visual outputs for technical and non-technical audiences.
5) Document methods, challenges, and evaluation in IEEE format.

II. Related Work

Housing affordability has been examined in policy studies, urban economics, and comparative institutional reports. However, literature often emphasizes either conceptual explanation or descriptive indicators, while providing limited implementation detail for reproducible city-level analytics systems.

Corrigan et al. [1] discuss Irish market dynamics in terms of supply frictions and persistent demand pressure. Their framing helps explain why prices may remain elevated despite policy interventions. Norris and Byrne [2] evaluate affordability within Irish policy transitions and social outcomes, but do not provide a full ETL architecture combining transaction data and semi-structured API indicators.

OECD comparative work [3] shows that affordability deterioration can occur even during periods of income growth when housing supply elasticity is weak. Glaeser and Gyourko [4] provide a theoretical basis for persistent divergence between housing prices and wages in constrained urban environments. Institutional Irish sources from CSO [5] and ESRI [6] provide high-quality indicators and policy framing, yet data delivery remains fragmented across formats and interfaces.

The main gap addressed in this project is the integration of high-volume transaction records with semi-structured API data in a reproducible dual-database pipeline tailored to Dublin.

III. Data Processing Methodology

A. Datasets

Three public datasets were selected:

- Property Price Register (CSV) [7]: transaction-level Dublin property prices.
- CSO Income API (JSON-stat) [5]: annual income indicators.
- CSO Population API (JSON-stat) [5]: contextual annual population indicators.

All selected sources are non-Kaggle public datasets and each source exceeds 1,000 records over extracted coverage.

B. ETL Architecture

Pipeline modules were organized as follows:

- `extract.py`: CSV loading, API retrieval, JSON-stat parsing.
- `transform.py`: cleaning, harmonization, metric engineering.
- `load.py`: PostgreSQL DDL and structured table loading.
- `main.py`: orchestration.
- `analyze.py`: static figure generation and insights export.

The processing flow follows: extract from source systems → persist raw data in MongoDB → transform and aggregate in Python → load curated outputs into PostgreSQL and processed CSV files.

C. Raw Storage and Lineage

MongoDB collections preserve extraction fidelity and traceability: `property_prices_raw`, `cso_income_raw`, `cso_income_parsed_raw`, `cso_population_raw`, and `cso_population_parsed_raw`. Storing both payload-level JSON and parsed rows enables reprocessing when upstream schema changes.

D. Cleaning and Transformation

Property data cleaning includes:

- column harmonization for inconsistent source headers,
- date parsing and year extraction,
- currency cleaning and numeric conversion,
- Dublin-only filtering, analysis year filtering (2015–2025),
- invalid/low-value outlier filtering.

Income and population API data are parsed from JSON-stat cubes into tabular format using dimension-order reconstruction. Time and value fields are normalized to numeric yearly series. Region and label variability are handled with fallback parsing logic.

E. Feature Engineering

Yearly indicators:

- `avg_price_eur`, `median_price_eur`, `sales_count`
- `affordability_index = avg_price_eur / income_eur`
- year-on-year growth metrics for price, income, and affordability
- optional `price_per_1000_residents`

Area-level aggregation uses robust area-label extraction:

1) Eircode prefix when available,
2) otherwise address-based extraction (e.g., Dublin district patterns),
3) fallback locality segment normalization.

Area groups with very low sample size are filtered (`transaction_count` $\geq$ 10) to reduce instability.

F. Processed Storage

Curated relational outputs are loaded into:

- `dublin_affordability_yearly`
- `dublin_affordability_area`
- `project_insights`

This schema supports consistent downstream analysis, dashboard serving, and reproducible reload behavior.

G. Implementation Challenges and Resolutions

Key engineering challenges and fixes:

- **PowerShell packaging parsing issue**: fixed by placing `param(...)` as the first executable statement.
- **Area collapse to unknown**: fixed by adding address-derived fallback area extraction.
- **Affordability range mismatch**: dashboard affordability chart constrained to valid income-overlap years.
- **PostgreSQL truncation on area labels**: schema widened from `VARCHAR(20)` to `VARCHAR(120)`.
- **API schema variability**: robust parser and region fallback logic added.

IV. Data Visualization Methodology

The visual design was guided by task-fit, readability, and interpretability for mixed audiences.

A. Chart-Type Justification

- **Line charts**: used for time trends (`avg_price_eur`, `income_eur`, affordability index), because slope and temporal continuity are central to interpretation.
- **Ranked bar chart**: used for top area comparison in a selected year; transaction count is shown in hover details for context.
- **Heatmap**: used to reveal persistent area-year structure and long-run spatial inequality.

B. Color and Interaction Design

Blue tones are used for baseline trend series and red for affordability stress to improve cognitive separation. Sequential palettes were chosen for magnitude encoding in heatmaps. The dashboard includes year selection and hover details to support exploratory analysis without overloading the static report visuals.

C. Dashboard Composition

The dashboard sequence is:

1) KPI cards,
2) city-level trend chart,
3) affordability trend chart,
4) year-selectable area ranking,
5) area-year heatmap,
6) textual key insights.

This layout provides both executive summary and analytical drill-down.

V. Results and Evaluation

A. Research Question Evaluation

The integrated series indicates that Dublin property prices increased substantially during 2015–2025. Over years where income overlap is available, affordability index values rise, indicating that housing required more income multiples over time. Therefore, the evidence supports the conclusion that affordability worsened.

B. Area-Level Findings

Area-level analysis shows that high-price zones remain persistently expensive rather than switching randomly year to year. This indicates structural spatial inequality in affordability burden across Dublin.

C. Key Insights

1) **Persistent divergence**: price growth outpaced income growth over overlap years.
2) **Affordability stress under activity**: transaction activity can remain high even when affordability worsens.
3) **Spatial persistence**: high-cost areas remain high-cost across years.

D. Figures

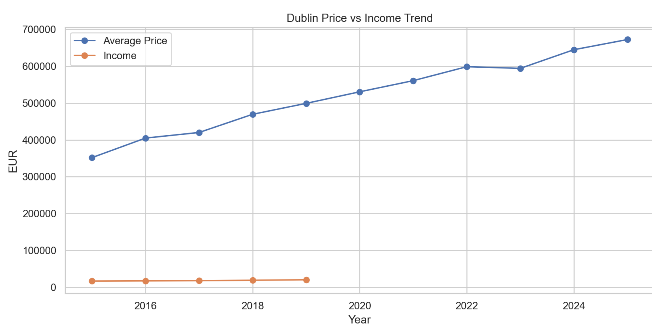

Fig. 1. Dublin property price versus income trend.

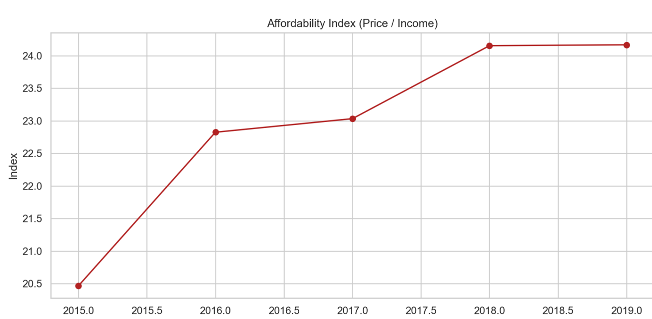

Fig. 2. Affordability index over years with valid income overlap.

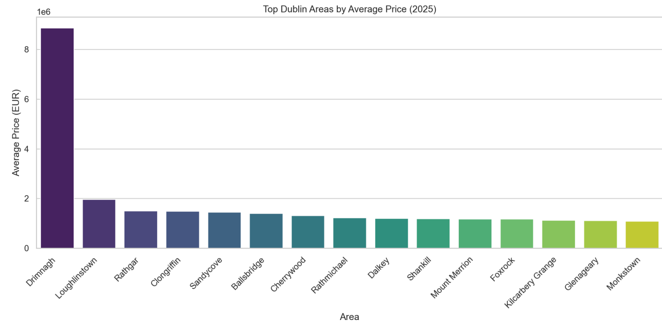

Fig. 3. Top Dublin areas by average price in the selected year.

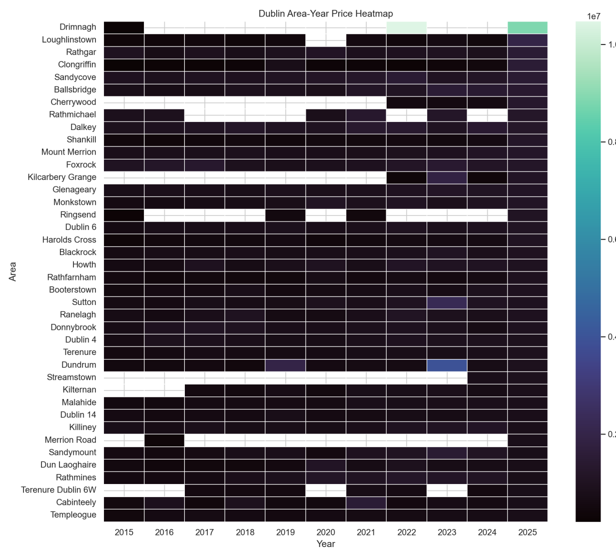

Fig. 4. Area-year heatmap of average property prices.

E. Critical Evaluation

This project meets the objective of an end-to-end analytics workflow, but several limitations remain. Income availability does not span all property years equally. Area derivation from addresses is practical yet approximate compared to formal geocoding boundaries. The baseline index is not inflation-adjusted and does not include mortgage interest burden. These limitations define interpretation boundaries and motivate future extensions.

VI. Conclusions and Future Work

This project delivered a reproducible, dual-database analytics system for evaluating Dublin housing affordability. The evidence indicates worsening affordability relative to income over available overlap years and persistent area-level inequality. The pipeline and dashboard demonstrate practical data engineering and communication capabilities aligned with module learning outcomes.

Future work includes inflation-adjusted and mortgage-aware affordability measures, distribution-aware income segmentation, geospatial enrichment with official boundary files, predictive modeling with uncertainty intervals, and orchestration/monitoring for scheduled pipeline refreshes.

Acknowledgment

The authors acknowledge National College of Ireland module guidance and public data availability from CSO and the Property Price Register.

References

[1] R. Corrigan, J. McQuinn, and C. O'Toole, "Housing market dynamics in Ireland: supply constraints and demand pressures," *Irish Economic Review*, vol. 50, no. 2, pp. 45–68, 2021.
[2] M. Norris and M. Byrne, "Housing affordability and policy transitions in Ireland," *Housing Studies*, vol. 33, no. 8, pp. 1201–1222, 2018.
[3] OECD, *Brick by Brick: Better Housing Policies*. Paris, France: OECD Publishing, 2022.
[4] E. L. Glaeser and J. Gyourko, "The economic implications of housing supply," *Journal of Economic Perspectives*, vol. 32, no. 1, pp. 3–30, 2018.
[5] Central Statistics Office, "PxStat API datasets and documentation," 2025. [Online]. Available: https://www.cso.ie/en/statistics/
[6] Economic and Social Research Institute, "Housing affordability and distributional pressures in Ireland," Dublin, Ireland, 2023.
[7] Government of Ireland, "Property Price Register," [Online]. Available: https://www.propertypriceregister.ie
[8] S. Few, *Show Me the Numbers: Designing Tables and Graphs to Enlighten*, 2nd ed. Burlingame, CA, USA: Analytics Press, 2012.
[9] C. Ware, *Information Visualization: Perception for Design*, 4th ed. Burlington, MA, USA: Morgan Kaufmann, 2021.